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① Find ∇f at the point $(4, 5, 0)$ if $f = y \log(x+y+z)$

Ans:-

$$f = y \log(x+y+z)$$

$$\nabla f = f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$$

$$= \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z}$$

$$= \hat{i} \left(\frac{y}{x+y+z} \right) + \hat{j} \left(\frac{y}{x+y+z} + \log(x+y+z) \right) + \hat{k} \left(\frac{y}{x+y+z} \right)$$

$$\therefore \nabla f_{(-4, 5, 0)} = \hat{i} \left(\frac{5}{-4+5+0} \right) + \hat{j} \left(\frac{5}{-4+5+0} + \log(-4+5+0) \right) + \hat{k} \left(\frac{5}{-4+5+0} \right)$$

$$= 5\hat{i} + 5\hat{j} + 5\hat{k}$$

② Find ∇W if $W = \log \sqrt{x^2+y^2+z^2}$

$$\nabla W = \hat{i} \frac{\partial W}{\partial x} + \hat{j} \frac{\partial W}{\partial y} + \hat{k} \frac{\partial W}{\partial z}$$

$$\frac{\partial W}{\partial x} = \frac{1}{\sqrt{x^2+y^2+z^2}} \cdot \frac{1}{2\sqrt{x^2+y^2+z^2}} \cdot 2x = \frac{x}{x^2+y^2+z^2}$$

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$$\frac{\partial W}{\partial y} = \frac{1}{\sqrt{x^2+y^2+z^2}} \cdot \frac{1}{2\sqrt{x^2+y^2+z^2}} \cdot 2y$$

$$= \frac{y}{x^2+y^2+z^2}$$

$$\frac{\partial W}{\partial z} = \frac{1}{\sqrt{x^2+y^2+z^2}} \cdot \frac{1}{2\sqrt{x^2+y^2+z^2}} \cdot 2z$$

$$= \frac{z}{x^2+y^2+z^2}$$

$$\nabla W = \left(\frac{x}{x^2+y^2+z^2} \right) \hat{i} + \left(\frac{y}{x^2+y^2+z^2} \right) \hat{j} + \left(\frac{z}{x^2+y^2+z^2} \right) \hat{k}$$

③ Find the unit normal ^{vectors} to the surface $yz + xz + xy = c$ at the point $(-1, 2, 3)$.

Ans:- $f = yz + xz + xy = c$.

$$\begin{aligned} \text{Normal to } f &= \nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} \\ &= \hat{i}(z+y) + \hat{j}(z+x) + \hat{k}(y+x) \\ \text{Now } \nabla f_{(-1, 2, 3)} &= \hat{i}(3+2) + \hat{j}(3-1) + \hat{k}(2-1) \\ &= 5\hat{i} + 2\hat{j} + \hat{k} \end{aligned}$$

$$\text{Unit normal vector} = \frac{\nabla f}{\|\nabla f\|} = \frac{5\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{25+4+1}} = \frac{5\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{30}}$$

① Let $f(x, y) = xy$. Find $D_u f(1, 2)$ for the unit vector $u = \frac{\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j}$.

Ans:- Given $f(x, y) = xy$

The partial derivatives of f are

$$f_x(x, y) = \frac{\partial f(x, y)}{\partial x} = y \Rightarrow f_x(1, 2) = 2$$

$$f_y(x, y) = \frac{\partial f(x, y)}{\partial y} = x \Rightarrow f_y(1, 2) = 1$$

$$\text{Now } D_u f(x_0, y_0) = f_x(x_0, y_0) u_1 + f_y(x_0, y_0) u_2$$

$$\Rightarrow D_u f(1, 2) = f_x(1, 2) u_1 + f_y(1, 2) u_2$$

$$= 2 \times \frac{\sqrt{3}}{2} + 1 \times \frac{1}{2}$$

$$= \underline{\underline{\sqrt{3} + \frac{1}{2}}} = \underline{\underline{2.23}}$$

$$u = \frac{\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j}$$

$$\Rightarrow u_1 = \frac{\sqrt{3}}{2}$$

$$u_2 = \frac{1}{2}$$

② If $f(x, y, z) = xy^2 + yz^3$, find $D_u f(2, -1, 1)$ in the direction of $\hat{i} + 2\hat{j} + 2\hat{k}$.

Ans:- Given $f(x, y, z) = xy^2 + yz^3$

The partial derivatives are

$$f_x = y^2, \quad f_y = 2yx + z^3, \quad f_z = 3yz^2$$

$$\Rightarrow f_x(2, -1, 1) = \underline{1}, \quad f_y(2, -1, 1) = 2 \times -1 \times 1 + 1^3 = \underline{-3}$$

$$f_z(2, -1, 1) = 3 \times -1 \times 1^2 = \underline{-3}$$

$$D_u f(x_0, y_0, z_0) = f_x(x_0, y_0, z_0)u_1 + f_y(x_0, y_0, z_0)u_2 + f_z(x_0, y_0, z_0)u_3$$

①

Here the given vector is not a unit vector.

$$\therefore \hat{u} = \frac{\hat{i} + 2\hat{j} + 2\hat{k}}{\sqrt{1^2 + 2^2 + 2^2}} = \frac{\hat{i} + 2\hat{j} + 2\hat{k}}{3} = \frac{1}{3}\hat{i} + \frac{2}{3}\hat{j} + \frac{2}{3}\hat{k}$$

$\therefore$ ① $\Rightarrow$

$$D_u f(2, -1, 1) = f_x(2, -1, 1)u_1 + f_y(2, -1, 1)u_2 + f_z(2, -1, 1)u_3 \\ = 1 \times \frac{1}{3} + -3 \times \frac{2}{3} + -3 \times \frac{2}{3} = \underline{\underline{-\frac{11}{3}}}$$

③ If $f = \frac{x}{x+y}$, find directional derivative at $P(1,0)$ in the direction of $Q(-1,-1)$.

Ans :- Here $\vec{u} = \frac{\vec{PQ}}{|\vec{PQ}|}$

$$\text{Here } \vec{PQ} = \vec{OQ} - \vec{OP}$$

$$= (-\hat{i} - \hat{j}) - (\hat{i} + 0\hat{j})$$

$$= -2\hat{i} - \hat{j}$$

$$|\vec{PQ}| = \sqrt{(-2)^2 + (-1)^2} = \sqrt{5}$$

$$\therefore \hat{u} = \frac{-2\hat{i} - \hat{j}}{\sqrt{5}} = -\frac{2}{\sqrt{5}}\hat{i} - \frac{1}{\sqrt{5}}\hat{j}$$

$$\Rightarrow \boxed{u_1 = -\frac{2}{\sqrt{5}}} \text{ and } \boxed{u_2 = -\frac{1}{\sqrt{5}}}$$

$$\text{Now } f(x,y) = \frac{x}{x+y}$$

$$\Rightarrow f_x = \frac{(x+y) \cdot 1 - x(1)}{(x+y)^2} = \frac{x+y-x}{(x+y)^2} = \frac{y}{(x+y)^2}$$

$$\Rightarrow \boxed{f_x(1,0) = 0}$$

$$f_y = \frac{(x+y) \cdot 0 - x(1)}{(x+y)^2} = \frac{-x}{(x+y)^2}$$

$$\Rightarrow \boxed{f_y(1,0) = -1}$$

$$\therefore D_u f(1,0) = 0 \times -\frac{2}{\sqrt{5}} + (-1) \times \frac{-1}{\sqrt{5}} = \underline{\underline{1/\sqrt{5}}}$$

④ $f = x^2 y z + 4 z^2 x$ along $2\hat{i} - \hat{j} - \hat{k}$
 at $(1, -2, -1)$. Find Directional derivative
 in the direction of $2\hat{i} - \hat{j} - \hat{k}$.
Ans:-

$$f = x^2 y z + 4 z^2 x$$

$$\Rightarrow f_x = 2xyz + 4z^2 \Rightarrow f_x(1, -2, -1) = 8$$

$$f_y = x^2 z \Rightarrow f_y(1, -2, -1) = -1$$

$$f_z = x^2 y + 8zx \Rightarrow f_z(1, -2, -1) = -10$$

$$\hat{u} = \frac{2\hat{i} - \hat{j} - \hat{k}}{\sqrt{2^2 + (-1)^2 + (-1)^2}} = \frac{2\hat{i} - \hat{j} - \hat{k}}{\sqrt{6}} = \frac{2}{\sqrt{6}}\hat{i} - \frac{1}{\sqrt{6}}\hat{j} - \frac{1}{\sqrt{6}}\hat{k}$$

$$D_u f(1, -2, -1) = f_x(1, -2, -1)u_1 + f_y(1, -2, -1)u_2 + f_z(1, -2, -1)u_3$$

$$= 8 \times \frac{2}{\sqrt{6}} + (-1) \times \frac{-1}{\sqrt{6}} + (-10) \times \frac{-1}{\sqrt{6}}$$

$$= \frac{16 + 1 + 10}{\sqrt{6}} = \underline{\underline{\frac{27}{\sqrt{6}}}}$$

② Find the directional derivative of $f(x, y) = e^{xy}$ at $(-2, 0)$ in the direction of the unit vector that makes the angle of $\pi/3$ with the positive x -axis.

Ans:- Given $f(x, y) = e^{xy}$

The partial derivatives of f are

$$f_x(x, y) = ye^{xy} \Rightarrow f_x(-2, 0) = 0.$$

$$f_y(x, y) = xe^{xy} \Rightarrow f_y(-2, 0) = -2.$$

The unit vector $\hat{u}$ that makes an angle of $\pi/3$ with the positive x -axis

$$\text{as } \hat{u} = \cos(\pi/3)\hat{i} + \sin(\pi/3)\hat{j}$$

$$= \frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}$$

$$\therefore D_{\hat{u}} f(x_0, y_0) = D_{\hat{u}} f(-2, 0)$$

$$= f_x(-2, 0)\cos \pi/3 + f_y(-2, 0)\sin \pi/3$$

$$= 0(1/2) + (-2)(\sqrt{3}/2) = \underline{\underline{-\sqrt{3}}}$$

Note:- A unit vector $\hat{u}$ in the xy -plane can be expressed as $\hat{u} = \cos \phi \hat{i} + \sin \phi \hat{j}$ where ϕ is the angle from the x -axis to $\hat{u}$. Formula is $D_{\hat{u}} f(x_0, y_0) = f_x(x_0, y_0)\cos \phi + f_y(x_0, y_0)\sin \phi$.

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③ Find the directional derivative of

$f(x, y, z) = x^2y - yz^3 + z$ at the point $(1, -2, 0)$ in the direction of the vector

$$\vec{a} = 2\hat{i} + \hat{j} - 2\hat{k}.$$

Ans:- We have $f(x, y, z) = x^2y - yz^3 + z$

The partial derivatives of f are

$$f_x(x, y, z) = 2xy, \quad f_y(x, y, z) = x^2 - z^3$$

$$f_z(x, y, z) = -3yz + 1$$

$$\Rightarrow f_x(1, -2, 0) = -4, \quad f_y(1, -2, 0) = 1, \quad f_z(1, -2, 0) = 1$$

Since $\vec{a}$ is not a unit vector, we normalise $\vec{a}$.

$$\therefore \hat{u} = \frac{\vec{a}}{\|\vec{a}\|} = \frac{2\hat{i} + \hat{j} - 2\hat{k}}{\sqrt{2^2 + 1^2 + (-2)^2}} = \frac{2\hat{i} + \hat{j} - 2\hat{k}}{\sqrt{9}} \\ = \frac{2}{3}\hat{i} + \frac{1}{3}\hat{j} - \frac{2}{3}\hat{k}$$

$$\text{Now } D_u f(x_0, y_0, z_0) = f_x(x_0, y_0, z_0)u_1 + f_y(x_0, y_0, z_0)u_2 + f_z(x_0, y_0, z_0)u_3$$

$$\therefore D_u f(1, -2, 0) = (-4)\left(\frac{2}{3}\right) + (1)\left(\frac{1}{3}\right) + (1)\left(-\frac{2}{3}\right) \\ = \underline{\underline{-3}}$$

✓ ① Find the directional derivative of $f = e^{2xy}$ at the point $(5,0)$ along

$$\vec{a} = -\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$$

Ans:- Given $f = e^{2xy}$, $\vec{a} = -\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$

$$\nabla f = f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$$

$$= \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z}$$

$$= \hat{i} (e^{2xy} \cdot 2y) + \hat{j} (e^{2xy} \cdot 2x)$$

$$(\nabla f)_{(5,0)} = \hat{i} (e^{2 \times 5 \times 0} \cdot 2 \times 0) + \hat{j} (e^{2 \times 5 \times 0} \cdot 2 \times 5)$$

$$= 0 + 10\hat{j} = \underline{\underline{10\hat{j}}}$$

Now $D_{\vec{a}} f(x_0, y_0) = [\nabla f(x_0, y_0)] \cdot \frac{\vec{a}}{\|\vec{a}\|} \rightarrow \text{①}$

$$\text{Now } \|\vec{a}\| = \sqrt{\left(-\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = 1$$

$$\therefore \frac{\vec{a}}{\|\vec{a}\|} = -\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$$

$$\therefore \text{①} \Rightarrow D_{\vec{a}} f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \frac{\vec{a}}{\|\vec{a}\|}$$

$$\begin{aligned} \Rightarrow D_{\vec{a}} f(5,0) &= (\nabla f)_{(5,0)} \cdot \frac{\vec{a}}{\|\vec{a}\|} \\ &= (10\hat{j}) \cdot \left(-\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}\right) \\ &= \underline{\underline{\frac{40}{5} = 8}} \end{aligned}$$

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② Find the directional derivative of $f = x^3 y^2 z^5 - 2xz + yz + 3x$ at the point $(-1, -2, 1)$ in the direction of $-ve z$ -axis.

Ans: Given $f = x^3 y^2 z^5 - 2xz + yz + 3x$

and $\vec{a} = -ve z\text{-axis} = -\hat{k}$

$$D_{\vec{a}} f(x_0, y_0, z_0) = \nabla f(x_0, y_0, z_0) \cdot \frac{\vec{a}}{\|\vec{a}\|} \rightarrow \text{①}$$

$$\begin{aligned} \text{Now } \nabla f &= f_x \hat{i} + f_y \hat{j} + f_z \hat{k} \\ &= [(3x^2 y^2 z^5) - 2z + 3] \hat{i} + \\ &\quad + (2x^3 y z^5 + z) \hat{j} + (5x^3 y^2 z^4 - 2x + y) \hat{k} \end{aligned}$$

$$\begin{aligned} \text{① } \nabla f(-1, -2, 1) &= [3 \times (-1)^2 (-2)^2 (1)^5 - 2 \times 1 + 3] \hat{i} + \\ &\quad + [2 \times (-1)^3 (-2) (1)^5 + 1] \hat{j} + [5 \times (-1)^3 (-2)^2 (1)^4 - 2 \times (-1) + (-2)] \hat{k} \\ &= 13 \hat{i} + 5 \hat{j} - 20 \hat{k} \end{aligned}$$

$$\text{①} \Rightarrow D_{\vec{a}} f(-1, -2, 1) = \nabla f(-1, -2, 1) \cdot \frac{-\hat{k}}{\sqrt{1}}$$

$$= (13\hat{i} + 5\hat{j} - 20\hat{k}) \cdot (-\hat{k})$$

$$= 20$$

$$(13\hat{i} + 5\hat{j} - 20\hat{k}) \cdot (0\hat{i} + 0\hat{j} + 1\hat{k}) = 20$$

③ Find the directional derivative of $f = e^{xy}$ at the point $(-2, 0)$ in the direction of the unit vector that makes an angle of $\pi/3$ with the +ve x-axis. Also find maximum value of D.D at $(-2, 0)$.

Ans.:- $f = e^{xy}$

$$D_{\vec{a}} f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \frac{\vec{a}}{\|\vec{a}\|} \quad \text{--- (1)}$$

$$\nabla f = f_x \hat{i} + f_y \hat{j}$$

$$= (ye^{xy}) \hat{i} + (xe^{xy}) \hat{j}$$

$$\nabla f(-2, 0) = (0 \times e^{-2 \times 0}) \hat{i} + (-2 \times e^{-2 \times 0}) \hat{j}$$

$$= -2 \hat{j}$$

Now the unit vector that makes an angle θ with the positive axis

$$\vec{a} = (\cos \theta) \hat{i} + (\sin \theta) \hat{j}$$

Here $\theta = \pi/3$

$$\therefore \vec{a} = (\cos \pi/3) \hat{i} + (\sin \pi/3) \hat{j}$$

$$= \frac{1}{2} \hat{i} + \frac{\sqrt{3}}{2} \hat{j}$$

$$\|\vec{a}\| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} = 1$$

$$\begin{aligned} \therefore \textcircled{1} \Rightarrow D_{\vec{a}} f(-2, 0) &= \nabla f(-2, 0) \cdot \frac{\vec{a}}{\|\vec{a}\|} \\ &= (-2\hat{j}) \cdot \left(\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j} \right) \\ &= \frac{-2\sqrt{3}}{2} = \underline{\underline{-\sqrt{3}}} \end{aligned}$$

$$\begin{aligned} \text{Maximum value of } Df(x_0, y_0) &= \|\nabla f\|_{(-2, 0)} \\ &= \|-2\hat{j}\| = \sqrt{(-2)^2} = \underline{\underline{2}} \end{aligned}$$

④ Find the directional derivative of $f = e^x \sec y$ at the point $P(0, \pi/4)$ in the direction of PQ where Q is the origin.

Ans:- Here $P(0, \pi/4)$ and $Q(0, 0)$

$$\begin{aligned} \therefore \vec{PQ} &= (0-0)\hat{i} + (0-\pi/4)\hat{j} \\ &= -\frac{\pi}{4}\hat{j} \end{aligned}$$

$$\therefore \vec{a} = \frac{(-\pi/4)\hat{j}}{\|(-\pi/4)\hat{j}\|} = -\hat{j}$$

$$D_{\vec{a}} f(0, \pi/4) = \left[\nabla f(0, \pi/4) \right] \cdot \frac{\vec{a}}{\|\vec{a}\|}$$

Given $f = e^x \sec y$

$$\nabla f = f_x \hat{i} + f_y \hat{j} = (e^x \sec y)\hat{i} + (e^x \sec y \tan y)\hat{j}$$

$$\nabla f(0, \pi/4) = \hat{i} (e^0 \sec \pi/4) + \hat{j} (e^0 \sec \pi/4 \tan \pi/4) \\ = \sqrt{2} \hat{i} + \sqrt{2} \hat{j} \quad \text{--- } ||\vec{a}||$$

$$||\vec{a}|| = ||-\pi/4 \hat{j}|| = \sqrt{(-\pi/4)^2} = \frac{\pi}{4}$$

$$D_{\vec{a}} f(0, \pi/4) = (\sqrt{2} \hat{i} + \sqrt{2} \hat{j}) \cdot \left(\frac{-\pi/4 \hat{j}}{\pi/4} \right) \\ = \frac{-\sqrt{2} (\pi/4)}{\pi/4} = -\sqrt{2}$$

Q5 Find the directional derivative of $f = xe^y - ye^x$ at $(0,0)$ along $\hat{i} - 2\hat{j}$

Q6 Find the directional derivative of $f = xe^y$ at $(1,1)$ along $\hat{i} + \hat{j}$.

Ans:-

Q5 Given $f = xe^y - ye^x$ at $(0,0)$

$$D_{\vec{a}} f(0,0) = [\nabla f(0,0)] \cdot \frac{\vec{a}}{||\vec{a}||}$$

$$\nabla f = f_x \hat{i} + f_y \hat{j}$$

$$= \hat{i} [e^y - ye^x] + [xe^y - e^x] \hat{j}$$

$$\Delta f(0,0) = \hat{i} - \hat{j}$$

$$\vec{a} = 5\hat{i} - 2\hat{j}$$

$$\|\vec{a}\| = \sqrt{5^2 + (-2)^2} = \sqrt{29}$$

$$D_{\vec{a}} f(0,0) = \nabla f(0,0) \cdot \frac{\vec{a}}{\|\vec{a}\|}$$

$$= (\hat{i} - \hat{j}) \cdot \left(\frac{5\hat{i} - 2\hat{j}}{\sqrt{29}} \right)$$

$$= \frac{5 + 2}{\sqrt{29}} = \frac{7}{\sqrt{29}}$$

⑥ Given $f = xe^y$ at $(1,1)$

$$D_{\vec{a}} f(1,1) = \nabla f(1,1) \cdot \frac{\vec{a}}{\|\vec{a}\|}$$

$$\nabla f = f_x \hat{i} + f_y \hat{j}$$

$$= (e^y) \hat{i} + (xe^y) \hat{j}$$

$$\nabla f(1,1) = e \hat{i} + e \hat{j} = e\hat{i} + e\hat{j}$$

$$\vec{a} = \hat{i} - \hat{j} \Rightarrow \|\vec{a}\| = \sqrt{2}$$

$$D_{\vec{a}} f(1,1) = \nabla f(1,1) \cdot \frac{\vec{a}}{\|\vec{a}\|}$$

$$= (e\hat{i} + e\hat{j}) \cdot \frac{(\hat{i} - \hat{j})}{\sqrt{2}} = \frac{e - e}{\sqrt{2}} = \underline{\underline{0}}$$

⑦ Find a unit vector in the direction in which $f = 3x - \log y$ increases more rapidly at the point $(2, 4)$. Also find the rate of change of f at the point $(2, 4)$ in that direction.

Ans:- Given $f = 3x - \log y$

We know that f increases more rapidly in the direction of ∇f .

$$\nabla f = f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$$

$$= 3 \hat{i} + \left(-\frac{1}{y}\right) \hat{j} =$$

$$\nabla f(2, 4) = 3 \hat{i} - \frac{1}{4} \hat{j}$$

Required unit vector $= \frac{\nabla f}{\|\nabla f\|} = \frac{3 \hat{i} - \frac{1}{4} \hat{j}}{\sqrt{9 + \frac{1}{16}}}$

$$= \frac{3 \hat{i} - \frac{1}{4} \hat{j}}{\frac{\sqrt{145}}{4}} = \frac{4(3 \hat{i} - \frac{1}{4} \hat{j})}{\sqrt{145}}$$

Rate of change of the function at $(2, 4)$

$$= \|\nabla f\| = \left\| 3 \hat{i} - \frac{1}{4} \hat{j} \right\|$$

$$= \sqrt{9 + \frac{1}{16}} = \frac{\sqrt{145}}{4}$$

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⑧ Find a unit vector in the direction in which $f = 20 - x^2 - y^2$ decreases most rapidly at $P(-1, 3)$. Also find the rate of change of f at P in that direction.

Ans:- Given $f = 20 - x^2 - y^2$

We know that f decreases most rapidly in the direction of $-\nabla f$.

$$\text{Now } \nabla f = f_x \hat{i} + f_y \hat{j}$$

$$= (-2x) \hat{i} + (-2y) \hat{j}$$

$$\nabla f(-1, 3) = 2 \hat{i} - 6 \hat{j} - 1\hat{e} = (2, -6) \hat{e}$$

$$\text{Required unit vector} = \frac{-\nabla f}{\|\nabla f\|}$$

$$= \frac{-2\hat{i} + 6\hat{j}}{\sqrt{4 + 36}} = \frac{-2\hat{i} + 6\hat{j}}{\sqrt{40}}$$

$$\text{Rate of change of } f \text{ at } (-1, 3) = \|\nabla f\|$$

$$\|\nabla f\| = \sqrt{2^2 + (-6)^2} = \sqrt{40}$$

$$\frac{-2\hat{i} + 6\hat{j}}{\sqrt{40}} = \frac{1}{\sqrt{10}} + 3\hat{j} =$$

Q Find the unit vector normal to the surface $z = 4x^2y$ through $P(1, -2)$

Ans: $z = 4x^2y$, $P(1, -2)$

Unit normal vector $= \frac{\nabla f}{\|\nabla f\|}$

$$\nabla f = f_x \hat{i} + f_y \hat{j} + f_z \hat{k} = (8xy) \hat{i} + (4x^2) \hat{j}$$

$$\nabla f(1, -2) = -16 \hat{i} + 4 \hat{j}$$

$$\|\nabla f\| = \sqrt{(-16)^2 + 4^2} = \sqrt{256 + 16} = \sqrt{302} = 4\sqrt{17}$$

$$\text{Unit normal} = \frac{\nabla f}{\|\nabla f\|} = \frac{-16 \hat{i} + 4 \hat{j}}{4\sqrt{17}}$$

HW Q Find the gradient of $f = y^2 z \tan^3 x$ at the point $(\pi/4, -2, 1)$

Ans: $f = y^2 z \tan^3 x$

$$\nabla f = f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$$

$$= (y^2 z \cdot 3 \tan^2 x \cdot \sec^2 x) \hat{i} + (2yz \tan^3 x) \hat{j} + (y^2 \tan^3 x) \hat{k}$$

$$\nabla f\left(\frac{\pi}{4}, -2, 1\right) = ((-2)^2 \cdot 3 \tan^2(\pi/4) \sec^2 \pi/4) \hat{i} + (2(-2) \tan^3 \pi/4) \hat{j} + (-2)^2 (\tan^3 \pi/4) \hat{k}$$

$$= (12 \times 2) \hat{i} - 4 \hat{j} + 4 \hat{k}$$

⑩ Find a unit vector in the direction in which $f = \sqrt{x^2 + y^2}$ increases most rapidly at $(4, 3)$. Also find the rate of change of f at the point $(4, 3)$ in that direction.

Ans :- Given $f = \sqrt{x^2 + y^2}$ at $(4, 3)$

f increases most rapidly in the direction of ∇f .

$$\nabla f = f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$$

$$= \frac{1 \times 2x}{2\sqrt{x^2 + y^2}} \hat{i} + \frac{1 \times 2y}{2\sqrt{x^2 + y^2}} \hat{j}$$

$$= \frac{x}{\sqrt{x^2+y^2}} \hat{i} + \frac{y}{\sqrt{x^2+y^2}} \hat{j}$$

$$\nabla f(4,3) = \frac{4}{\sqrt{4^2+3^2}} \hat{i} + \frac{3}{\sqrt{4^2+3^2}} \hat{j}$$

$$= \frac{4}{5} \hat{i} + \frac{3}{5} \hat{j}$$

$$\|\nabla f\| = \sqrt{\left(\frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2} = \sqrt{\frac{16+9}{25}} = 1$$

$$\text{Unit vector} = \frac{\nabla f}{\|\nabla f\|} = \frac{4}{5} \hat{i} + \frac{3}{5} \hat{j}$$

Rate of change of function at (4,3)

$$= \|\nabla f\| = 1$$

(12) Find the unit vector in the direction in which the function f decreases most rapidly at P where $f = 4e^{xy} \cos z$ and also

find the rate of change of f at

$$P(0,1,\pi/4)$$

Ans :- $f = 4e^{xy} \cos z$, $P(0, 1, \pi/4)$

$$\nabla f = f_x \hat{i} + f_y \hat{j} + f_z \hat{k}$$

$$= (4e^{xy} \cos z \cdot y) \hat{i} + (4e^{xy} \cos z \cdot x) \hat{j} + (4e^{xy} (-\sin z)) \hat{k}$$

$$\nabla f_{(0, 1, \pi/4)} = (4e^0 \cos \pi/4) \hat{i} + (4e^0 \cos \pi/4 \cdot 0) \hat{j} + (4e^0 (-\sin \pi/4)) \hat{k}$$

$$= \frac{4}{\sqrt{2}} \hat{i} - 4 \hat{k} = 2\sqrt{2} \hat{i} - 2\sqrt{2} \hat{k}$$

Unit vector = $-\frac{\nabla f}{\|\nabla f\|} = -2\sqrt{2} \hat{i} + 2\sqrt{2} \hat{k}$

$$\frac{\left(\frac{4}{\sqrt{2}} + \frac{1}{\sqrt{2}}\right)}{\sqrt{\frac{16}{2} + 1}} = \frac{\|\nabla f\|}{\|\nabla f\|} = \frac{-2\sqrt{2} \hat{i} + 2\sqrt{2} \hat{k}}{\sqrt{8+8}}$$

$$= -\frac{\sqrt{2}}{2} \hat{i} + \frac{\sqrt{2}}{2} \hat{k}$$

$$= -\frac{1}{\sqrt{2}} \hat{i} + \frac{1}{\sqrt{2}} \hat{k}$$

Rate of change = $-\|\nabla f\| = -\sqrt{8+8}$
 $= -4$

Q13 Find the maximum value of D.D of

① $f = \frac{x}{z} + \frac{y}{z^2}$ at $(1, 2, 2)$ also find unit vector

② Find the minimum value of D.D of

$f = \frac{x+z}{z-y}$ at $(5, 2, 6)$ and unit vector in that direction.

Homework Answer

Qn Find the minimum value of the directional derivative of $f(x, y, z) = \frac{x+z}{z-y}$ at $(5, 7, 6)$. Also find the unit vector in that direction.

Ans:-

$$f(x, y, z) = \frac{x+z}{z-y}$$

$$f_x = \frac{1}{z-y} \Rightarrow f_x(5, 7, 6) = \frac{1}{6-7} = \underline{\underline{-1}}$$

$$f_y = (x+z) \left(\frac{-1}{(z-y)^2} \right) (-1)$$

$$\left[\frac{x+z}{(z-y)^2} \right] \Rightarrow f_y(5, 7, 6) = \frac{5+6}{(6-7)^2} = \underline{\underline{11}}$$

$$f_z = \frac{(z-y) \cdot 1 - (x+z) \cdot 1}{(z-y)^2}$$

$$= \frac{z-y-x-z}{(z-y)^2}$$

$$= \frac{-(y+x)}{(z-y)^2}$$

$$\Rightarrow f_z(5, 7, 6) = \frac{-(7+5)}{(6-7)^2} = \underline{\underline{-12}}$$

$$\nabla f(5, 7, 6) = f_x(5, 7, 6) \hat{i} + f_y(5, 7, 6) \hat{j} + f_z(5, 7, 6) \hat{k}$$

$$= -1 \hat{i} + 11 \hat{j} + 12 \hat{k}$$

$$= -\hat{i} + 11\hat{j} + 12\hat{k}$$

$$\text{Minimum value} = -\|\nabla f\| = -(5+10) = -15$$

$$= -\sqrt{1^2 + 11^2 + 12^2}$$

$$= -\sqrt{1 + 121 + 144}$$

$$= -\sqrt{266}$$

$$\text{Unit vector in the decreasing direction} = \frac{-\nabla f}{\|\nabla f\|}$$

$$= -\frac{\nabla f}{\|\nabla f\|}$$

$$= \frac{-(-\hat{i} + 11\hat{j} + 12\hat{k})}{\sqrt{266}}$$

$$= \frac{\hat{i} - 11\hat{j} - 12\hat{k}}{\sqrt{266}}$$

$$= \frac{\hat{i} - 11\hat{j} - 12\hat{k}}{\sqrt{266}}$$

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